

Is traditional text analysis dead in an era of LLMs?

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Related Papers

Inference for Regression with Variables Generated by AI or Machine Learning

(with Battaglia, Christensen, Sacher)

Macro Shocks and Firm-Level Response Heterogeneity

(with Davis and Seminario-Amez)

Policymakers' Uncertainty

(with Cieslak, McMahon, Xiao)

LLMs and Measurement Error

What is Traditional Text Analysis?

Dictionary/Pattern-Matching Methods

Dimensionality Reduction of Doc-Term Matrix (LSA, LDA, NMF)

Word Embeddings (GloVe, word2vec)

“Small” Transformer Models (BERT, RoBERTa, DistilBERT, GPT 1/2)

“Large” AI Models (ChatGPT, Claude)

Comparing Methods on Concrete Task

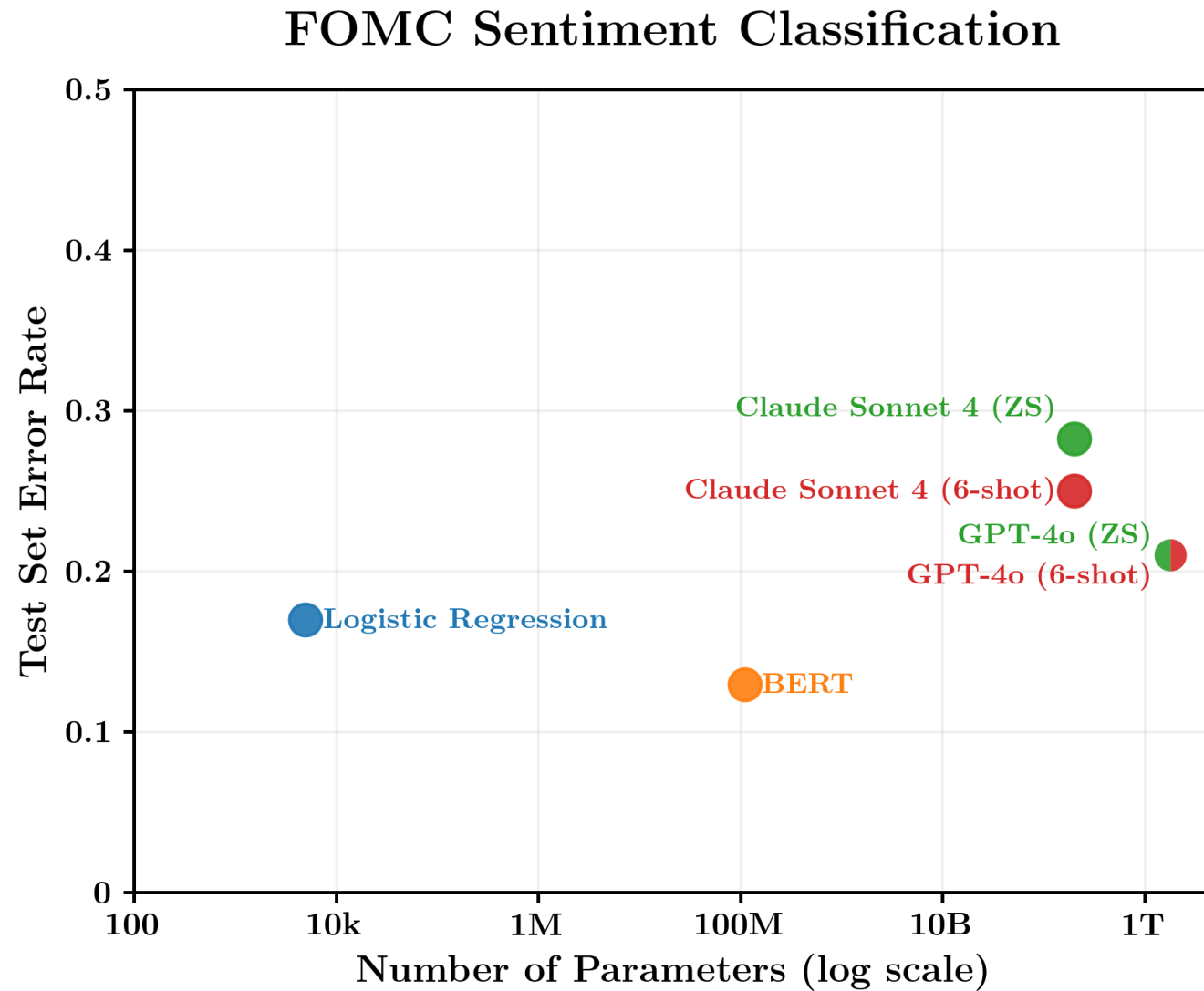
Gorodnichenko et. al. (2023) builds a measure of hawkish sentiment of FOMC statements

Training data is 1,243 human-labeled statement sentences: hawkish, neutral, dovish

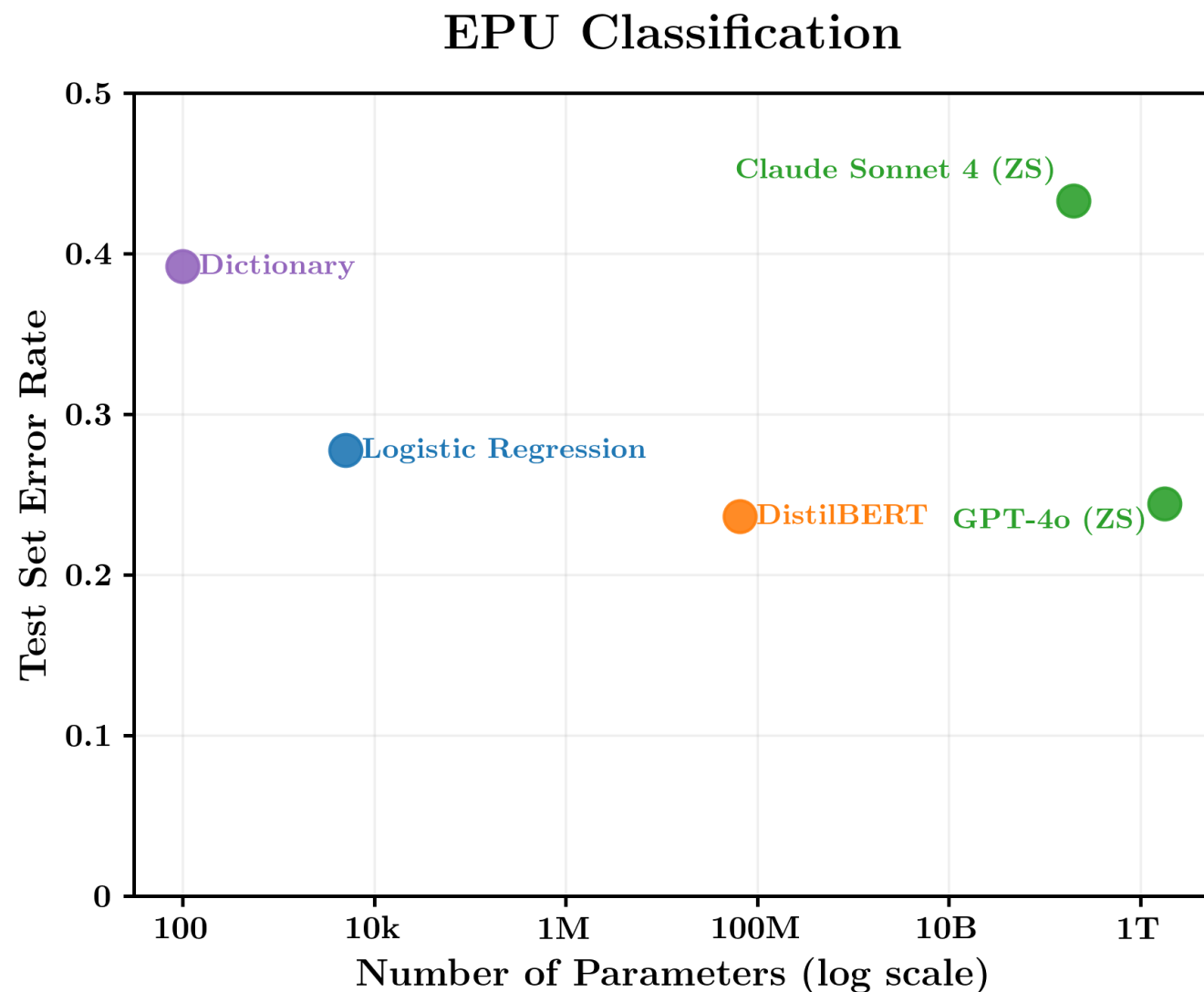
Split into 80/10/10 portions of training, validation, and test data

Compare test-set performance of a variety of methods

Results



Results for Economic Policy Uncertainty Labeling



Lessons

Material misclassification probability in standard economic measurement problems

Classification performance and model size have non-monotonic relationship

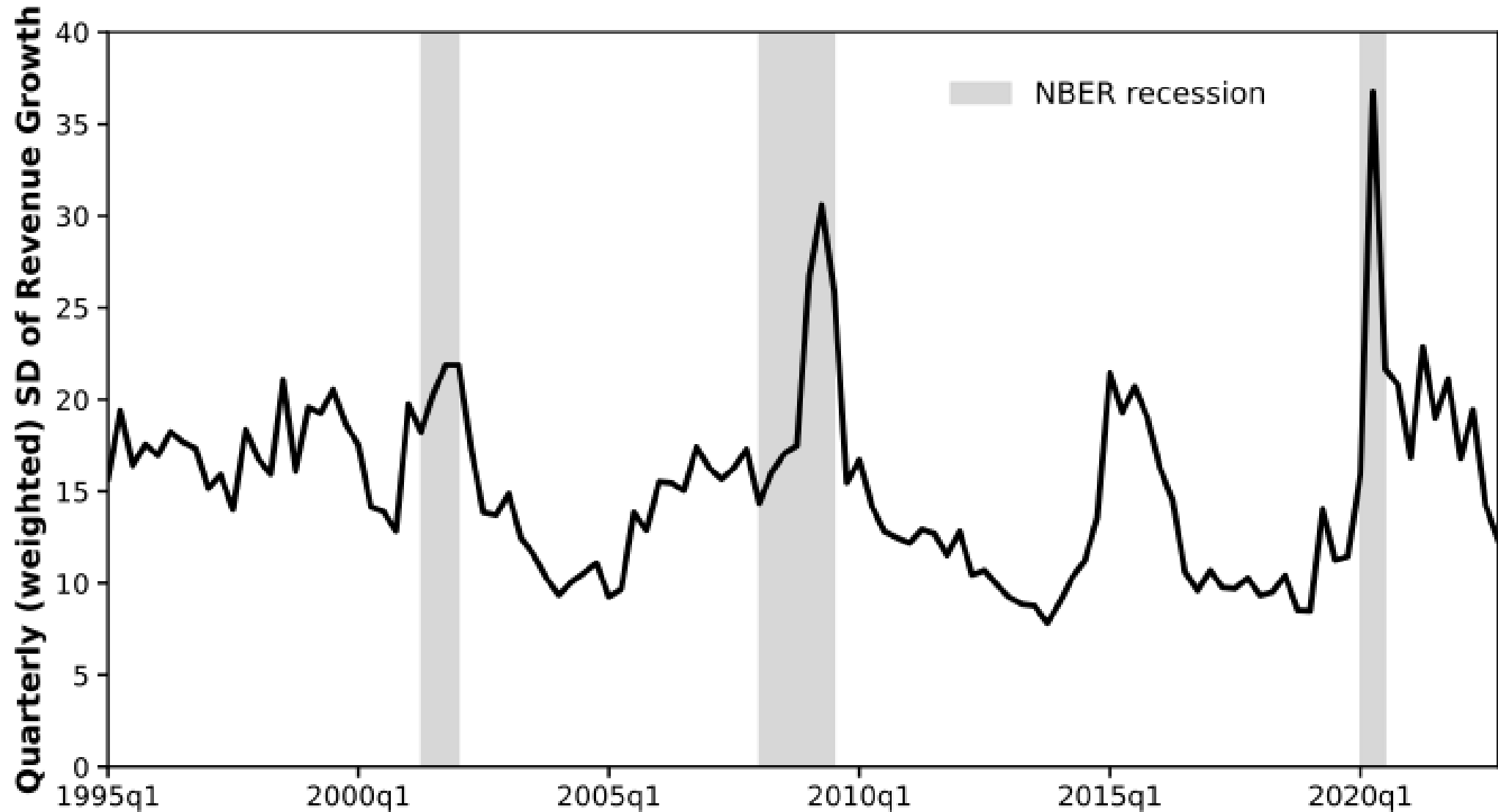
Modern AI models have not “solved” the problem of mapping text to variables

Small, fine-tuned LLMs are a more compelling choice **when training data exists**

Battaglia et al (2026) provides bias correction for regression onto AI-generated indices, implemented via Python package **ValidMLInference** and R package **MLBC**

Bag-of-Words to Denoise/Interpret Asset-Based Shocks

Dispersion in Firm-Level Outcomes is Countercyclical



Possible Explanations

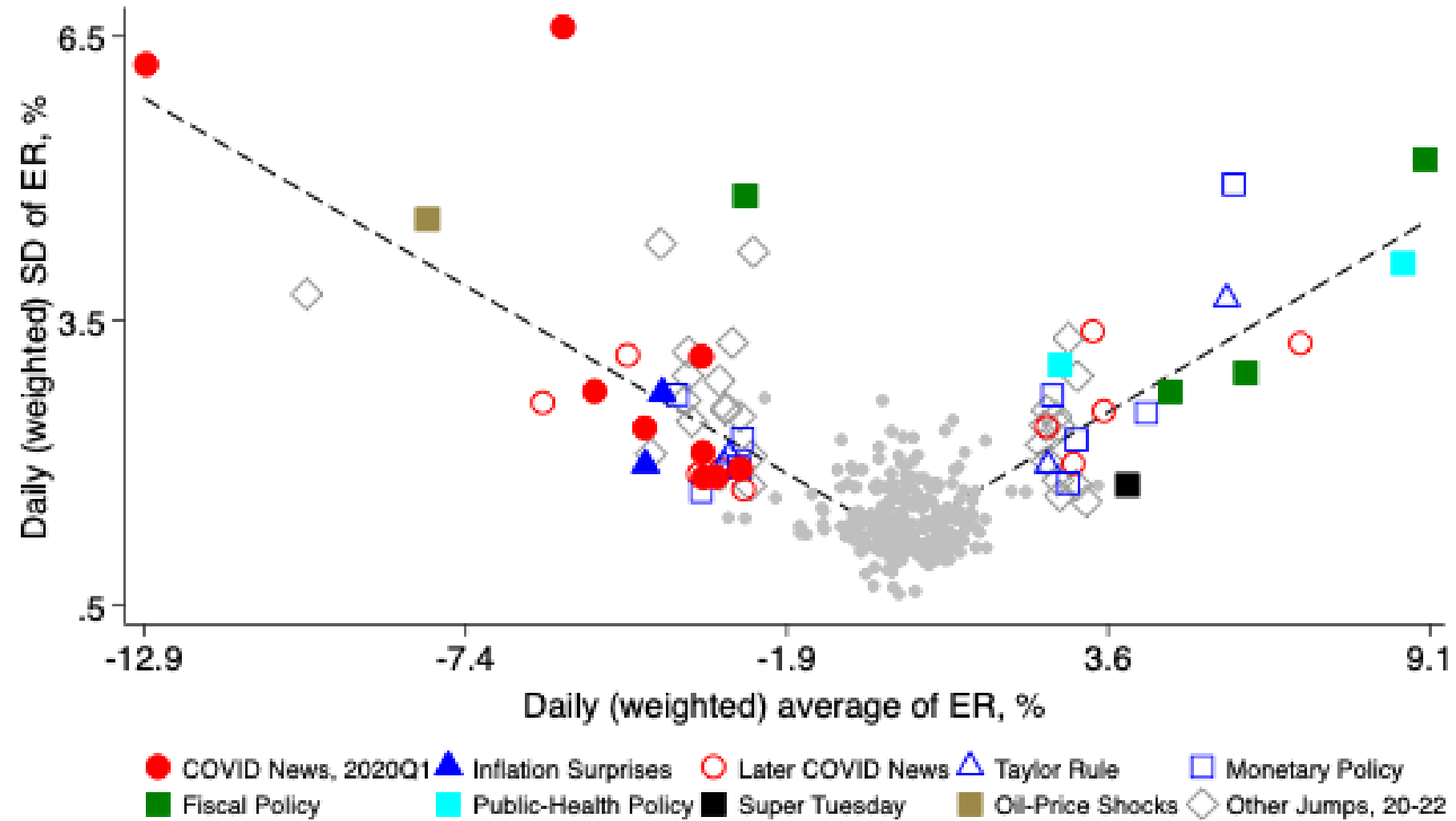
Leading explanation (Bloom 2009; Bloom et al 2018)

Firm output is $Y_{it} = Z_{it} X_t N_{it}^\alpha$ and variance of Z_{it} rises in recessions

Our explanation (Davis et al 2025)

Firm output is $Y_{it} = Z_{it} \prod_{s=1}^S (X_t^s)^{\theta_i^s} N_{it}^\alpha$ where X_t^s is the realization of shock type s in period t and θ_i^s is firm i 's exposure to type- s shocks

Evidence from Jump Dates



Baseline Empirical Specification

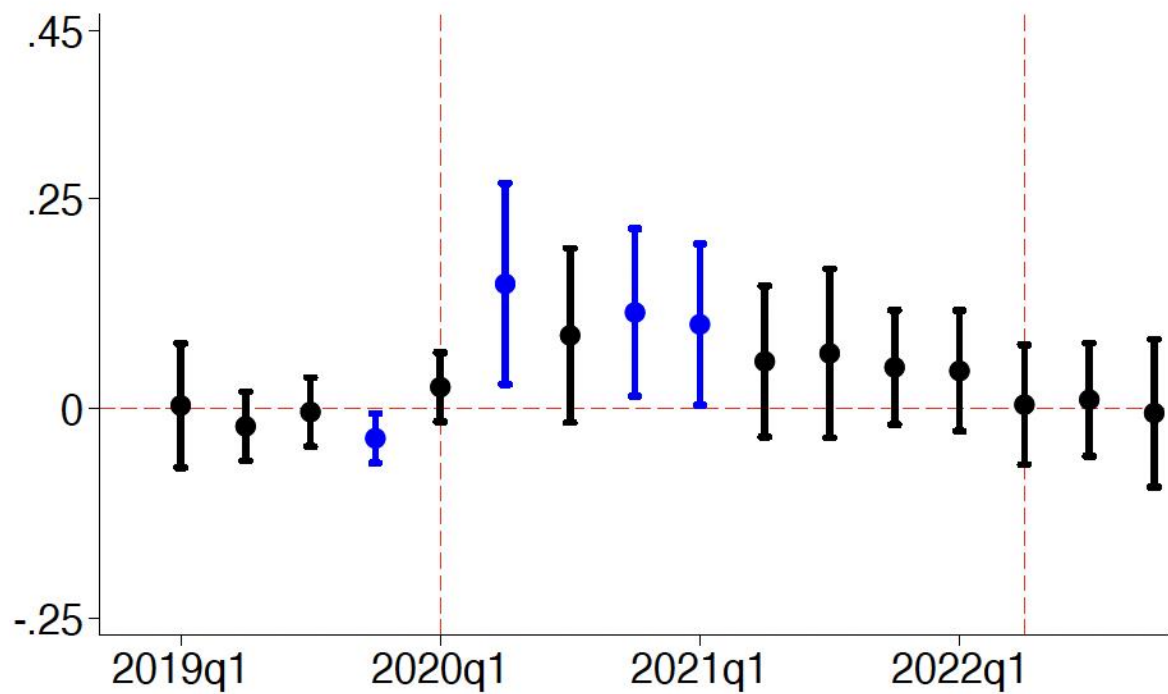
$$g_{it} = \mu_i + \mu_{s(i),t} + \alpha'_t \mathbf{e}_i + \beta'_t \mathbf{x}_{it} + \epsilon_{it}$$

- ▶ μ_i : firm fixed effects
- ▶ $\mu_{s(i),t}$: sector $\times$ time fixed effects
- ▶ $\mathbf{e}_i$: shock exposure measures (jump-date returns)
- ▶ α_t : time-varying coefficients of interest
- ▶ $\mathbf{x}_{it}$: controls (leverage, log assets)

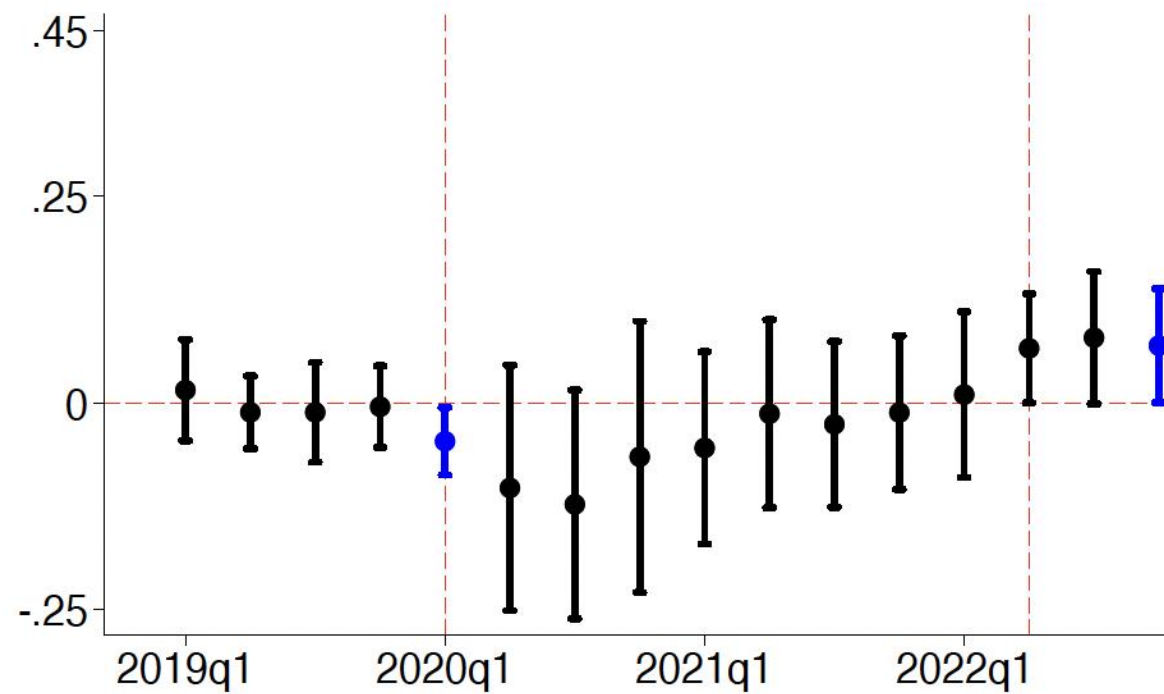
Outcome: $g_{it} = \log(\text{rev}_{i,t}) - \log(\text{rev}_{i,t-12})$ (12-quarter revenue growth).
Balanced panel of $\approx 1,000$ COMPUSTAT firms, 2015Q1–2022Q4.

Initial exposures: $\mathbf{e}_i = (R_i^P, R_i^I)$ where

- ▶ R_i^P : average abnormal return on 9 pandemic dates (2020Q1)
- ▶ R_i^I : average abnormal return on 2 inflation dates (2022)



Pandemic



Inflation

Figure: Revenue Growth and **Raw-Return** Exposures

Informative Text: Risk Factors from 10K Reports

To capture hard-to-measure firm characteristics driving heterogeneous exposure, we use the risk-factors discussion of 10-K regulatory filings.

These texts discuss factors that generate risk/uncertainty in future earnings; exhaustive due to their legal status.

Full set of RF corpora downloaded for the 2015-2019 time period. We use one compiled report by firm.

Multinomial Inverse Regression

Procedure from [Taddy \(2013\)](#) and applied in [Gentzkow et. al. \(2019\)](#).

Consider $\text{AbnRet}_i^{\text{P}}$, firm-level average return across 9 **pandemic** dates.

For each term v (14K in corpus), estimate Poisson regression

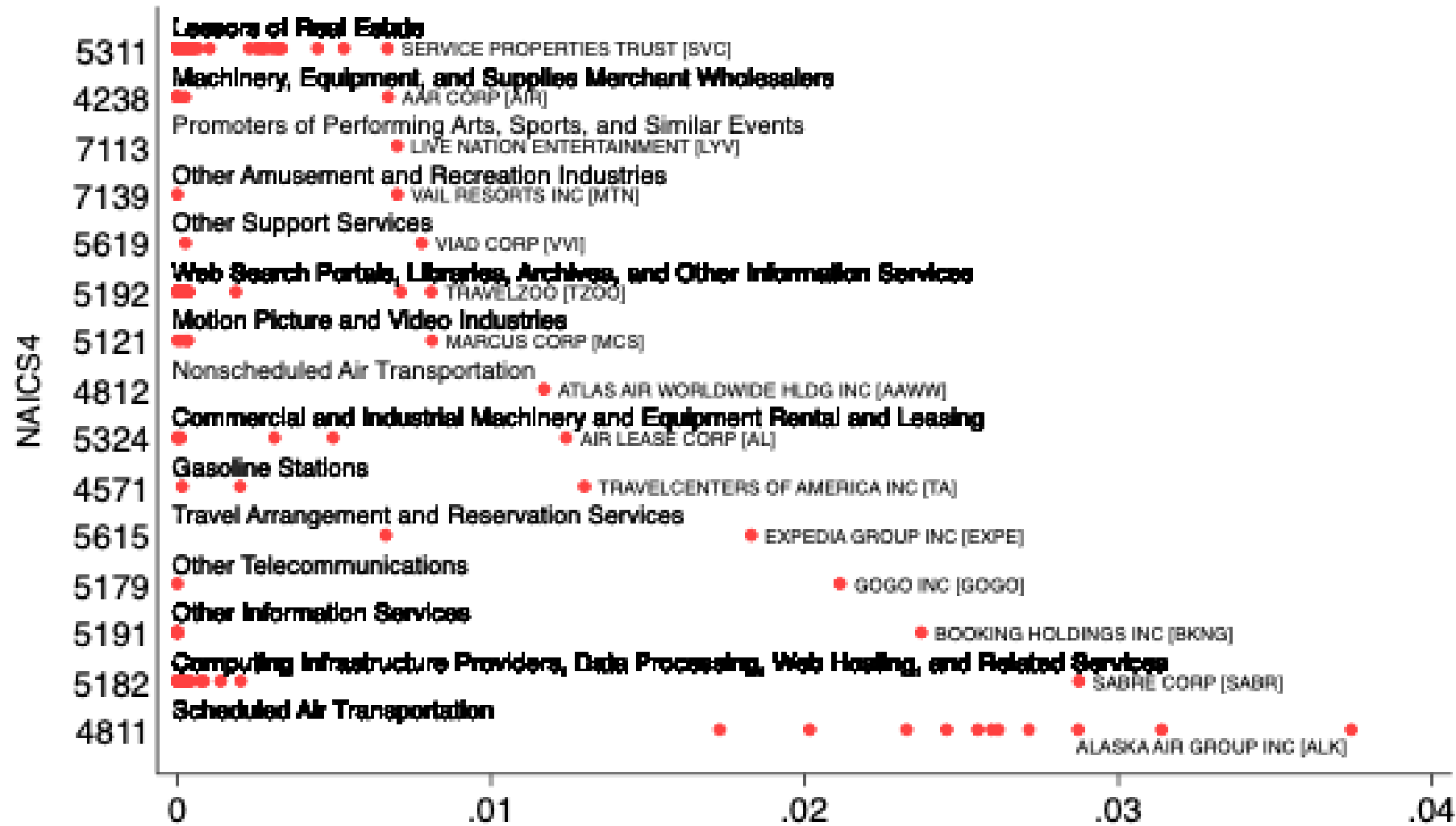
$$\log \mathbb{E}[c_{i,v}] = \alpha_v + \beta_v \text{AbnRet}_i^{\text{P}} + \theta'_v \text{controls}_i$$

- ▶ $c_{i,v}$ is count of term v in firm i 's text.
- ▶ controls_i : NAICS2 effects; and pre-shock leverage and firm size (market cap).

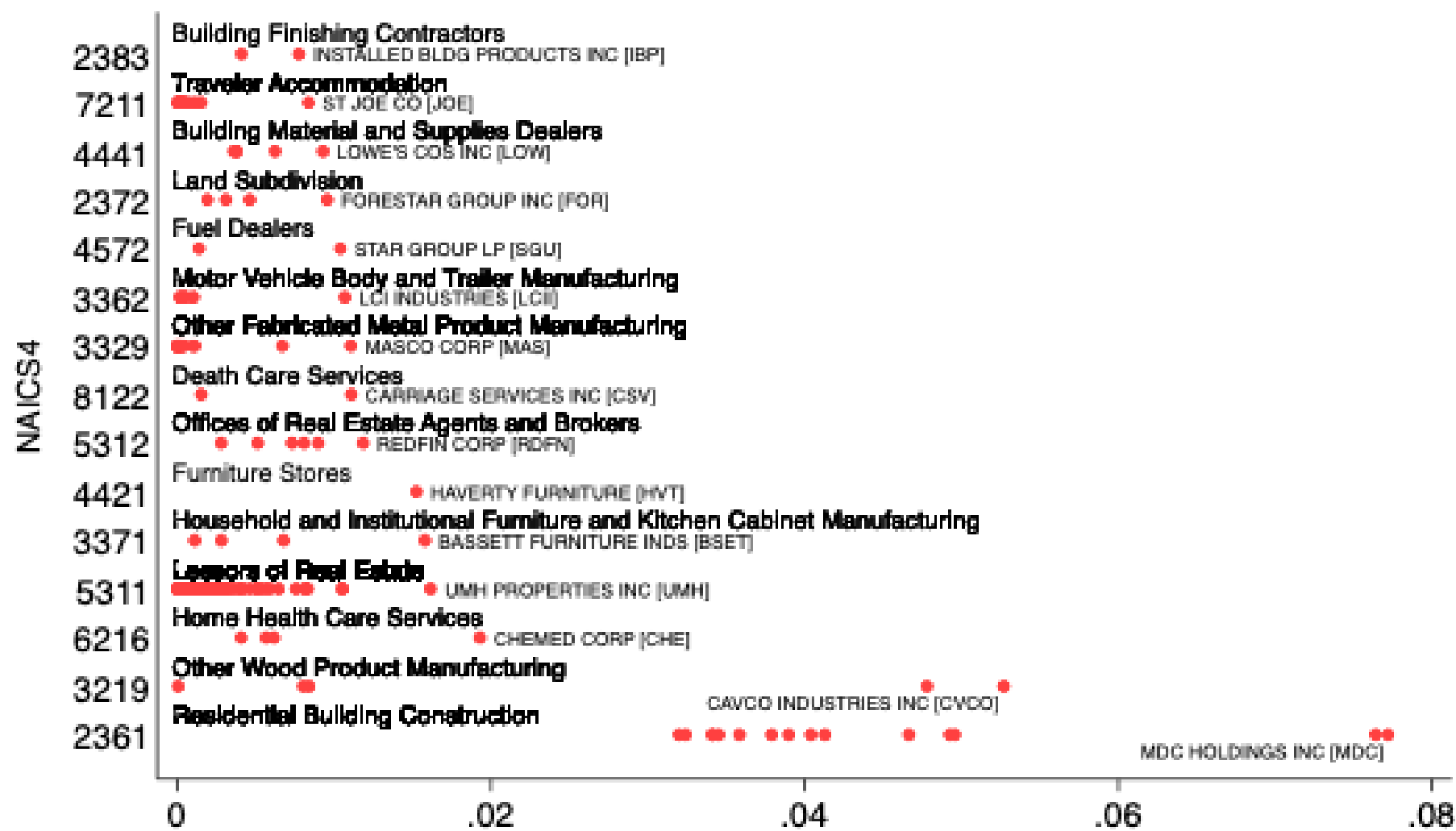
$z_i^{\text{P}} = \sum_v \hat{\beta}_v \frac{c_{i,v}}{C_i}$ is a weighted sum of firm i 's term counts (**pandemic** exposure)

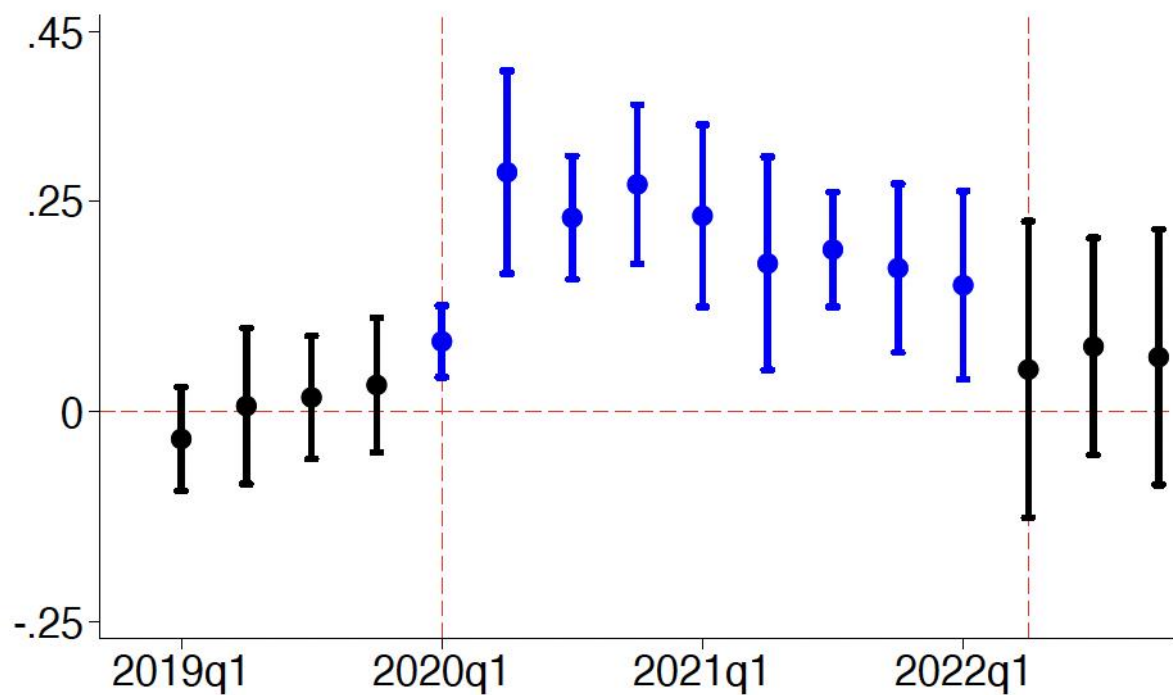
Repeat exercise for **inflation** dates, yielding z_i^{I} (**inflation** exposure).

Distribution of “Air Travel” Terms

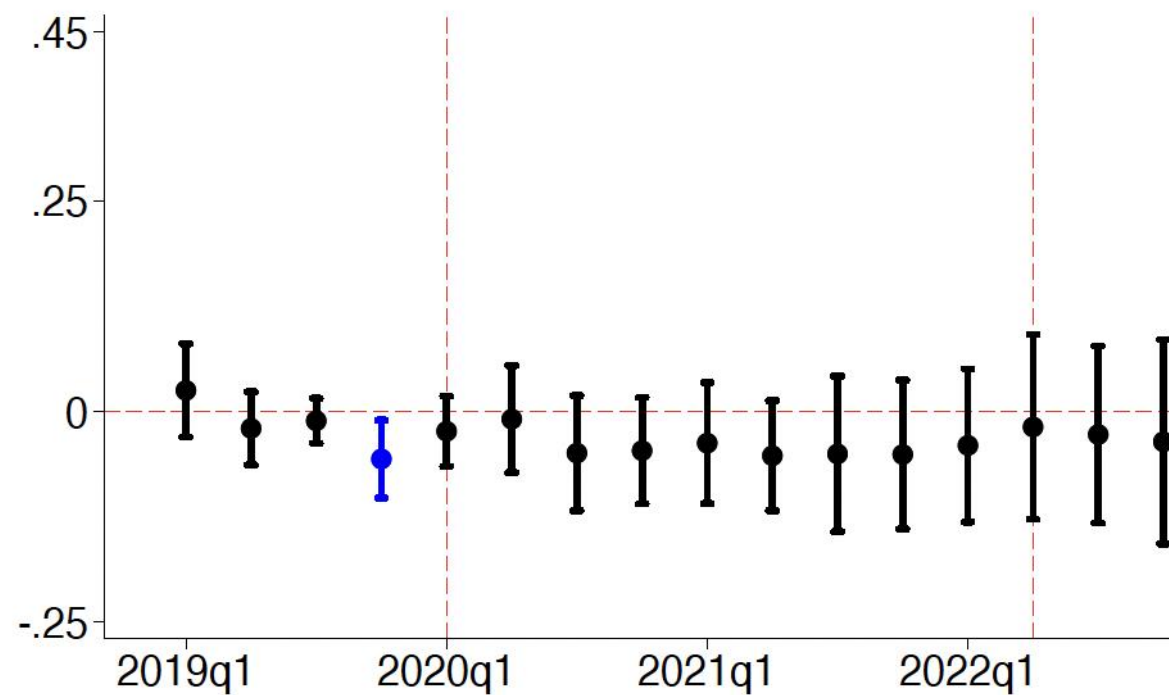


Distribution of “Housing” Terms



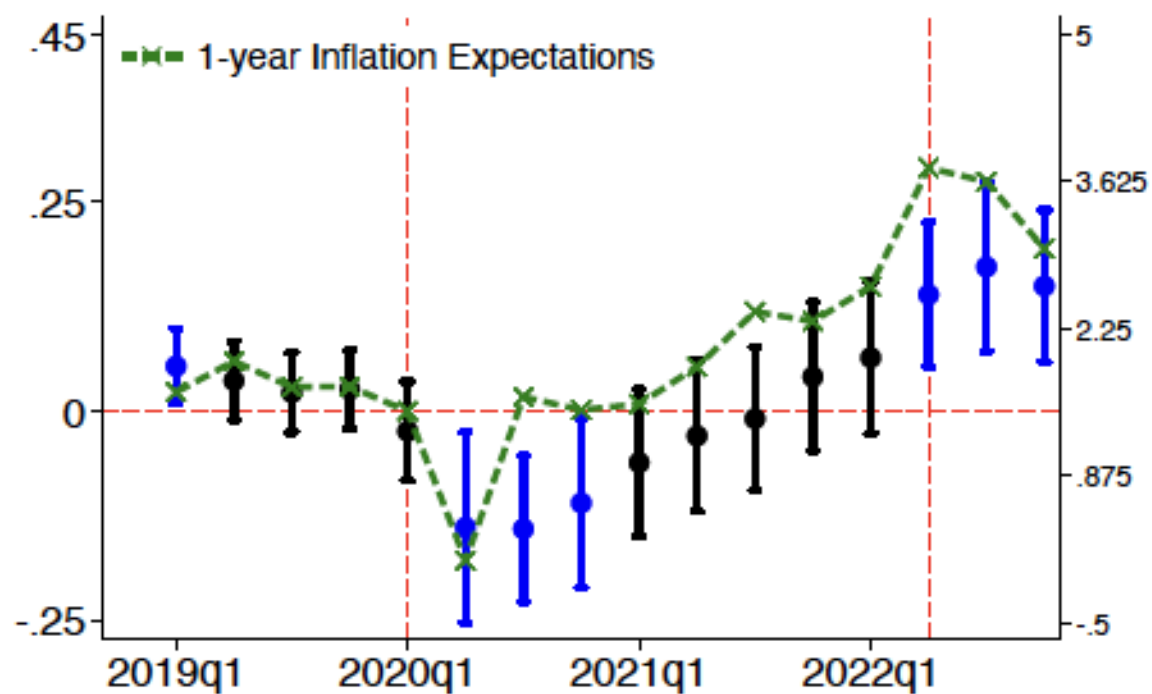


Text-based Pandemic

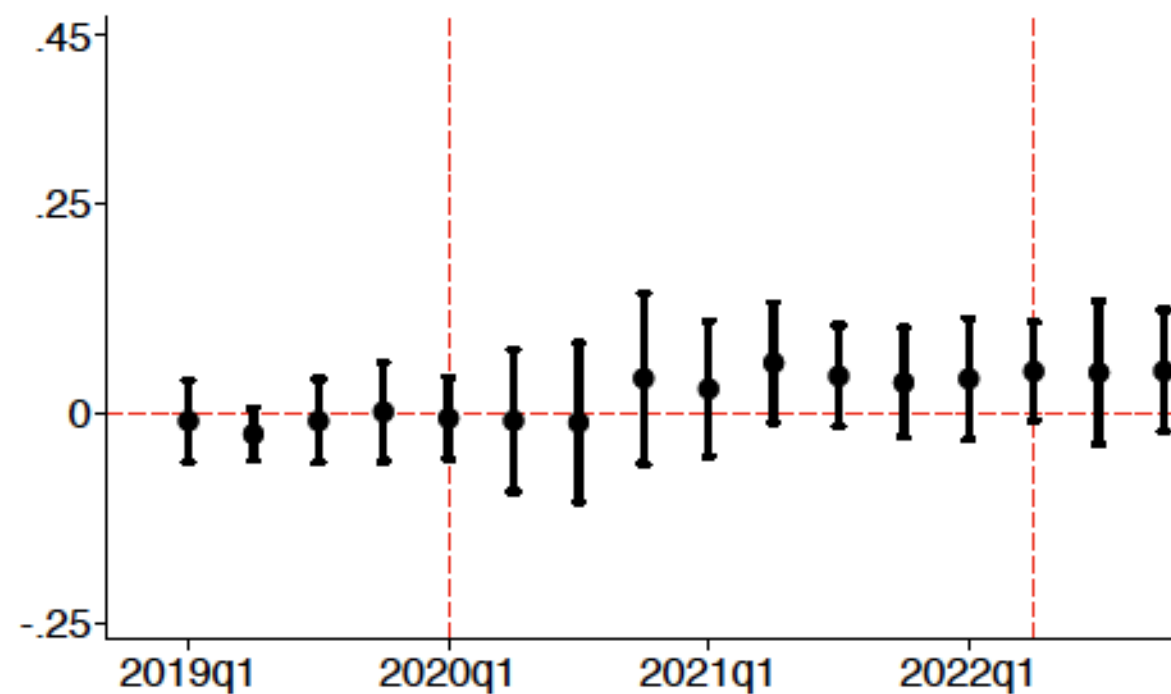


Residual Pandemic

Figure: Revenue Growth and **Pandemic** Exposures



Text-based Inflation



Residual Inflation

Figure: Revenue Growth and **Inflation** Exposures

LLMs and Narrative Analysis

Monetary Policy and Uncertainty

Policymakers have long viewed uncertainty as central to decision-making

*“Uncertainty is not just an important feature of the monetary policy landscape; it is the defining characteristic of that landscape...As a consequence, the conduct of **monetary policy** in the United States has come to involve, at its core, crucial elements of **risk management**.”*
(Greenspan 2004)

*“The **Governing Council** bases its monetary policy decisions, including the evaluation of the proportionality of its decisions and potential side effects, on an integrated assessment of all relevant factors. In particular, it **takes into account** not only the most likely path for inflation and the economy but also **surrounding risks and uncertainty**”* (ECB Strategy Statement 2025)

Uncertainty also discussed in 2025 **Fed’s strategy review** (Bauer 2025, Garga 2025)

Research Questions

Do policymakers respond to risk/uncertainty?

→ how to measure risk/uncertainty?

→ What are its policy effects?

If so, why?

→ Certainty equivalence vs insurance

FOMC Transcripts

FOMC transcripts are publicly available from Fed website from 1976 onwards, published with a five-year lag.

Verbatim, attributed, (nearly) complete record of policy deliberations.

We focus on scheduled meetings, not conference calls.

Main sample covers October 1982—December 2019, 299 meetings in total.

107 unique FOMC members.

Meeting Structure

FOMC meetings follow a regular structure in much of our sample. Core parts:

1. Discussion of Economic/Financial Conditions (FOMC1)
 1. Staff presentation on economic conditions
 2. Q&A on staff presentation
 3. FOMC member presentations on economic conditions
2. Discussion of Policy Response (FOMC2)
 1. Staff presentation on policy alternatives
 2. Q&A on policy alternatives
 3. FOMC members sequentially state and justify preferred alternative

Measuring Risk Perceptions at the Individual Level

We feed the text from FOMC1 for each meeting/member into an LLM (Claude Sonnet 4).

Separately for **inflation** and **growth**, we map the text into one of four risk categories:

- Upside
- Balanced
- Downside
- Not Discussed

$BoR_{it}^v \in \{U, B, D, N\}$ is member i 's coded view for variable $v \in \{I, G\}$ in meeting t .

Query focusses on risk perceptions around individual forecasts, not disagreement with staff.

LLM reasoning must include direct quotations from transcript.

Example Quotations (Inflation)

"We can't dismiss the possibility that compensation growth will drift upward, raising core inflation"

Yellen, May 1996

"I would also judge the risk to the inflation outlook as being biased to the downside"

Lockhart, Sep 2010

"The risks to the inflation outlook remain balanced"

Bostic, May 2019

Example Quotations (Growth)

*"I'm beginning to think **our greatest problem may be an excessive increase in activity**"*

Partee, Feb 1985

*"The **risk** to satisfactory **economic performance**... will remain skewed very much to the **downside**"*

Kohn, Mar 2008

*"**Considerable risks** still in place to **growth**, including gasoline prices, Europe and the deleveraging of European banks, continued stresses and tight credit in the U.S. banking system..."*

Bernanke, Mar 2012

Meeting-Level Balance-of-Risks Skew

Let $R_{D,t}^v = \sum_i 1(BoR_{it}^v = D)$ and $R_{U,t}^v = \sum_i 1(BoR_{it}^v = U)$ be the number of FOMC members in meeting t classified as expressing downside and upside risks about a variable v .

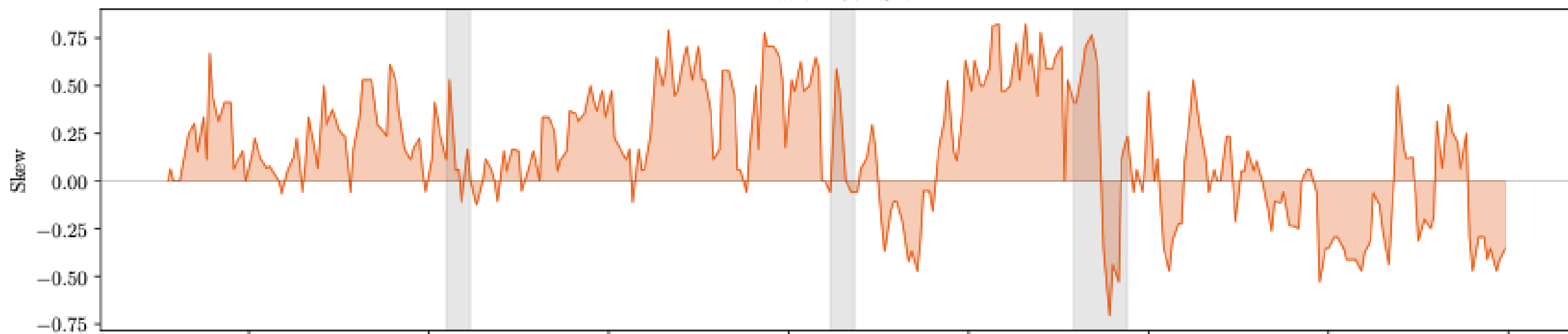
Our aggregate measure of risk is

$$BoR_t^v = \frac{R_{U,t}^v - R_{D,t}^v}{N_t}$$

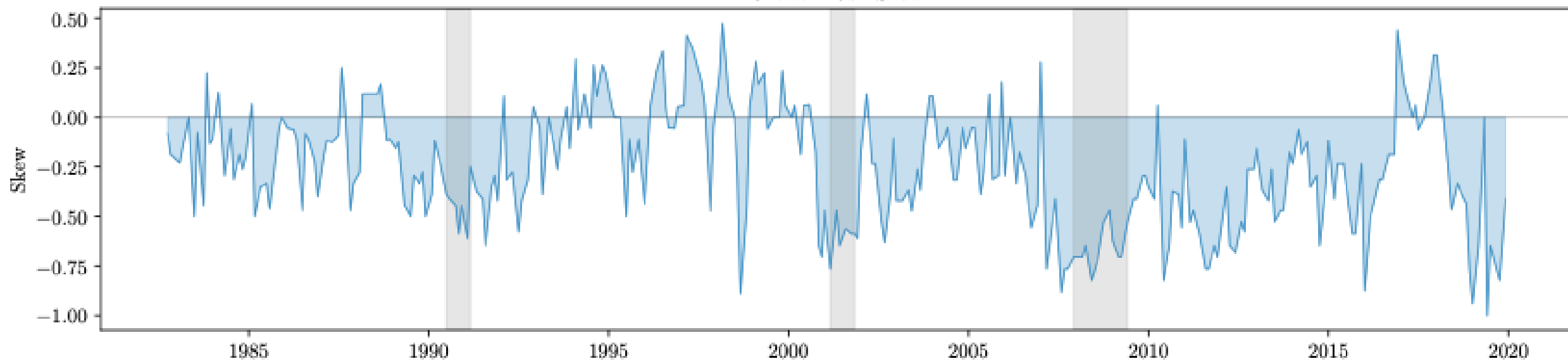
where N_t is the number of FOMC members in meeting t .

Captures skewness in distribution of individual risk perceptions in any given meeting.

Inflation Risk Skew



Growth Risk Skew



Validation I: Dictionary Methods

We also build dictionaries to measuring members' uncertainty/risk perceptions.

Policymakers' Uncertainty (PMU) indices for inflation/disinflation and growth, respectively, tabulate the frequency of risk/uncertainty-related terms in sentences with topical keywords.

Sentiment indices tabulate directional terms in sentences with topical keywords.

Advantage is simplicity, transparency, replicability.

Disadvantage is less sophisticated understanding of context.

	BoR Infl Skew				BoR Growth Skew		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
PMU: Inflation	0.214*** (0.015)			0.175*** (0.021)			
PMU: Deflation		-0.161*** (0.016)		-0.124*** (0.015)			
PMU: Growth					-0.131*** (0.019)		-0.116*** (0.017)
Sent: Inflation			0.205*** (0.018)	0.060*** (0.022)			
Sent: Growth						0.166*** (0.015)	0.155*** (0.014)
R2	0.396	0.226	0.369	0.606	0.176	0.289	0.428
Num.Obs.	258	258	258	258	258	258	258

HC1 robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Validation II: Summary of Economic Projections

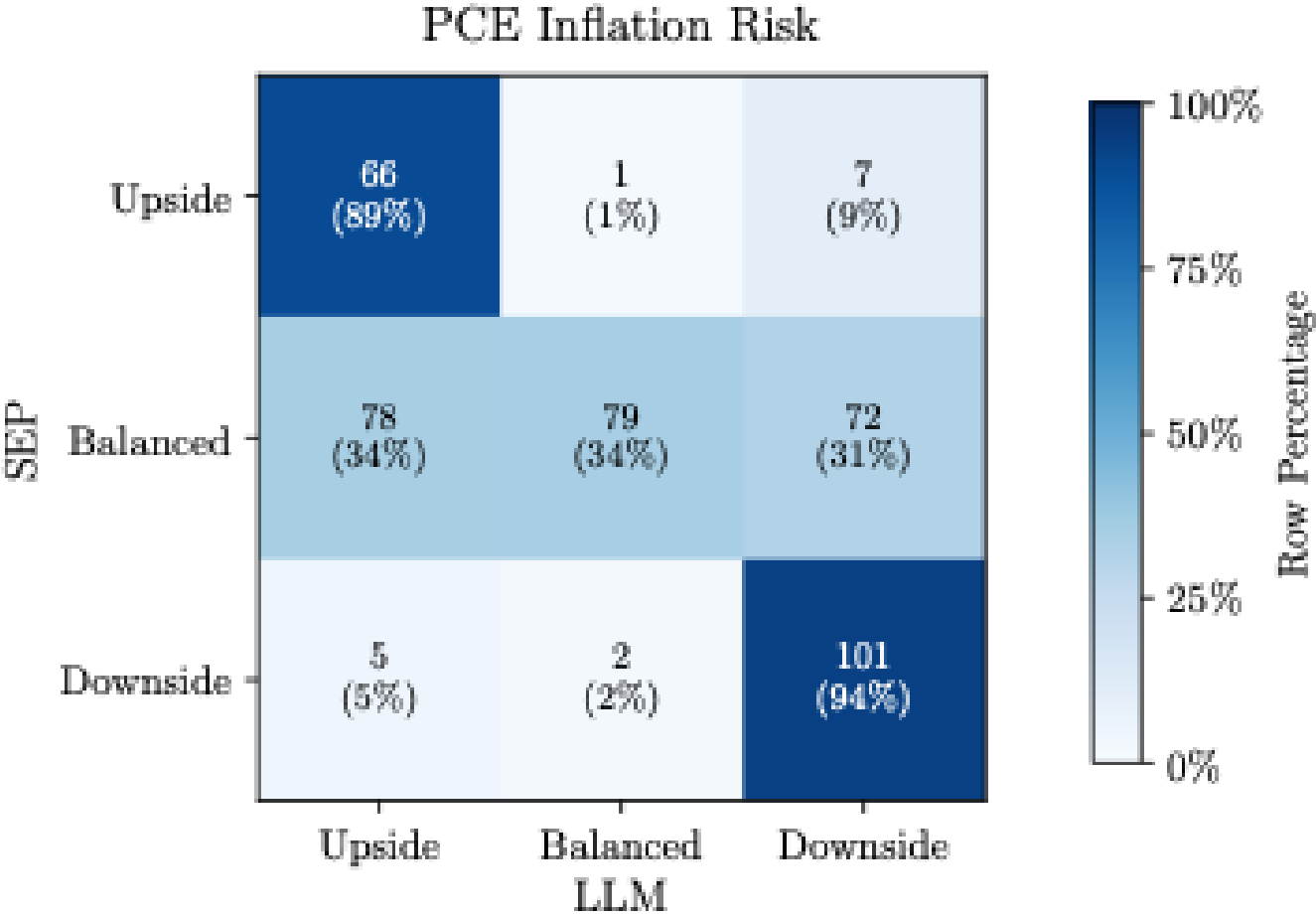
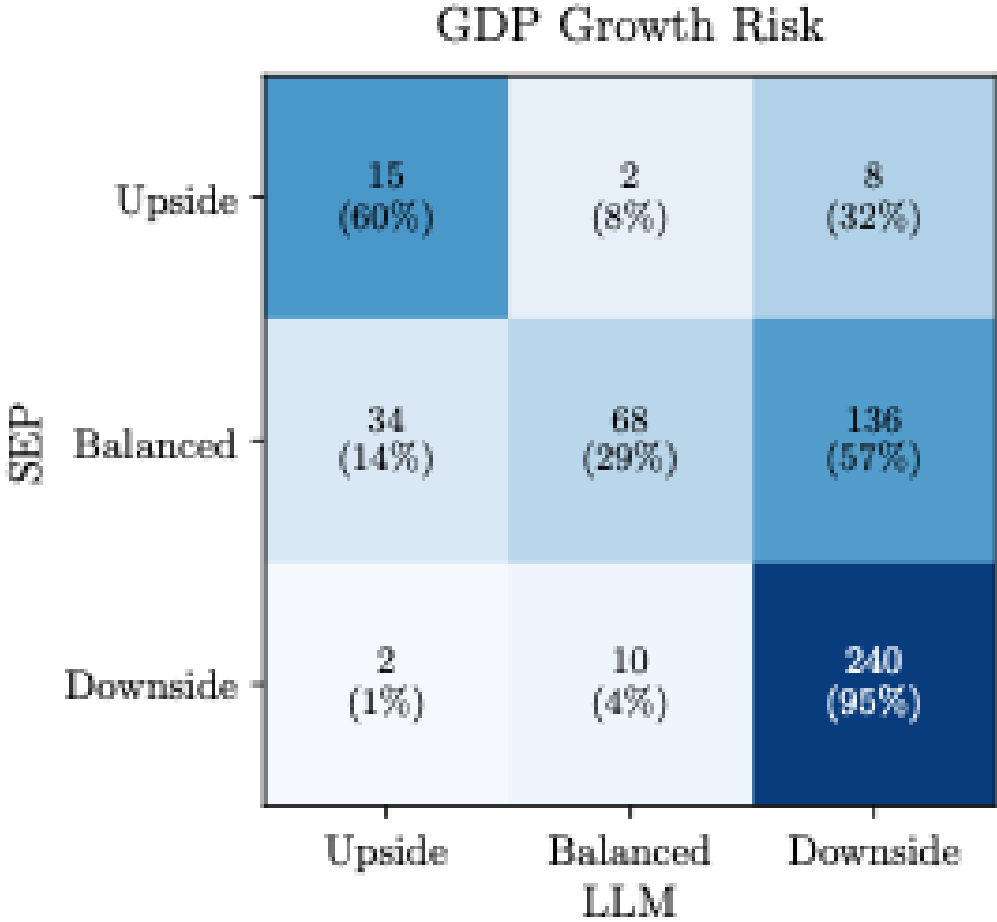
From Oct 2007, and four times a year from 2008 onwards, FOMC members respond to a survey about economic conditions collated in the [Summary of Economic Projections](#).

Includes question “*Please indicate your judgment of the risk weighting around your projections*” with three responses:

- Weighted to downside
- Broadly balanced
- Weighted to upside

Additional questions on member-specific point estimates of macro variables and optimal policy used as controls in regression analysis.

Comparison to SEP



Examples of Mismatch

Kocherlakota Apr 2012, Inflation, LLM: Upside, SEP: Balanced

“In this scenario, we might well see rapid increases in inflation and inflation expectations when labor markets seem weak by historical standards...This does not strike me as a risk that we should ignore.”

Yellen Dec 2013, Growth, LLM: Balanced, SEP: Downside

“I now see appreciably diminished downside risks to the outlook for the labor market and more balanced risks to growth than in September.”

Bullard Sep 2016, Inflation, LLM: Downside, SEP: Upside

“I do remain concerned that market-based measures of inflation expectations remain too low to be consistent with the 2 percent inflation forecast. This is a risk to the outlook that I have described.”

Monetary Policy Shocks

1. Romer and Romer shocks

→ orthogonalize Fed Funds Rate with respect to Greenbook controls.

2. Aruoba and Drechsel shocks:

→ further orthogonalize with respect to textual sentiment indicators extracted from staff materials (color books)

→ explicitly motivated by potential divergence between modal numeric Greenbook forecasts and conditional mean forecast, driven by asymmetric balance of risks

	RR Shock			AD Shock		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: All meetings with FFR target</i>						
BoR Inflation	0.200*** (0.043)		0.178*** (0.045)	0.051*** (0.013)		0.046*** (0.013)
BoR Growth		0.124** (0.052)	0.064 (0.054)		0.032** (0.015)	0.020 (0.014)
R2	0.085	0.033	0.093	0.064	0.026	0.073
Num.Obs.	201	201	201	207	207	207
<i>Panel B: Greenspan era onwards (1987-)</i>						
BoR Inflation	0.189*** (0.043)		0.155*** (0.045)	0.046*** (0.014)		0.040*** (0.014)
BoR Growth		0.147*** (0.054)	0.091 (0.057)		0.033** (0.015)	0.021 (0.015)
R2	0.090	0.056	0.108	0.064	0.034	0.077
Num.Obs.	163	163	163	169	169	169

Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Panel Regression Model

To more clearly isolate the relationship between balance of risks and policy preferences, we conduct member-level analysis by fitting the model

$$y_{it} = \alpha_i + \delta_t + \sum_{x \in \{U, B, N\}} \gamma_x^I 1(BoR_{it}^I = x) + \sum_{x \in \{U, B, N\}} \gamma_G^I 1(BoR_{it}^G = x) + \varepsilon_{it}$$

for a variety of preference measures y_{it} .

$\delta_t \rightarrow$ absorbs time trends, unobserved meeting-level shocks

$\alpha_i \rightarrow$ controls for fixed member preferences, e.g. BoR captures hawk/dove heterogeneity

Errors clustered at meeting level.

Preference Variables

Official voting record of the FOMC has little heterogeneity.

Meade (2005) shows that dissent in voice is more common than dissent in voting and builds a human-coded dataset of individual FFR preferences dataset covering 1989-1997.

Preferences: Voiced FFR preference from Meade (1989-1997)

	(1)	(2)	(3)	(4)	(5)	(6)
Infl Risk: Upside	6.366*** (1.276)		5.470*** (1.264)	3.448** (1.413)		3.247** (1.420)
Growth Risk: Upside		4.157*** (1.386)	3.213** (1.336)		1.024 (1.185)	0.578 (1.160)
Balanced/ND dummies	Y	Y	Y	Y	Y	Y
Meeting FE	Y	Y	Y	Y	Y	Y
Speaker FE	N	N	N	Y	Y	Y
Num.Obs.	1119	1119	1119	1119	1119	1119

Cluster-robust SEs (by meeting) in parentheses. Omitted category: downside risk. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

SEP Fed Funds Rate Forecast

Starting in 2012, the SEP added a question about members' preferences regarding the path of future interest rates. Most variation at one- and two-year-ahead horizons.

We match BoR_{it}^v to SEP responses at meeting-speaker level and use FFR_{it}^h as y_{it} .

We use SEP to further control for:

1. Individual growth/inflation forecasts (one-year-ahead GDP and core PCE forecasts)
2. Individual uncertainty around growth and inflation forecasts (relative to last 20 years)

Helps **isolate perceptions about higher moments** of distribution of economic variables reflected in BoR_{it}^v from those about first moments.

Preferences: SEP Future FFR Preferences (2012-2019)

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: FFR Forecast $t + 1$ (pp)</i>						
Infl Risk: Upside	0.543*** (0.089)		0.533*** (0.090)	0.201** (0.086)		0.201** (0.089)
Growth Risk: Upside		0.382*** (0.133)	0.017 (0.100)		0.103 (0.089)	-0.007 (0.077)
Num.Obs.	554	554	554	551	551	551
<i>Panel B: FFR Forecast $t + 2$ (pp)</i>						
Infl Risk: Upside	0.680*** (0.135)		0.690*** (0.133)	0.278** (0.127)		0.300** (0.130)
Growth Risk: Upside		0.560*** (0.188)	0.060 (0.159)		0.135 (0.136)	0.002 (0.120)
Num.Obs.	554	554	554	551	551	551
Balanced/ND dummies	Y	Y	Y	Y	Y	Y
Point forecast controls	Y	Y	Y	Y	Y	Y
SEP uncertainty controls	Y	Y	Y	Y	Y	Y
Meeting FE	Y	Y	Y	Y	Y	Y
Speaker FE	N	N	N	Y	Y	Y

Cluster-robust SEs (by meeting) in parentheses. Omitted category: downside risk. Point forecast controls: member’s own SEP forecast ($t + 1$). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Insurance at the Fed

Recall that BoR can affect policy via shifting expected economic conditions or insurance.

Used by Blinder and Reis (2004) to characterize elements of Greenspan era.

*“By cutting interest rates to offset the negative effects of financial turmoil on aggregate economic activity, monetary policy can **reduce the likelihood** that a financial disruption might set off an adverse feedback loop...monetary policy cannot—and should not—aim at minimizing valuation risk, but policy should aim at reducing macroeconomic risk.”* (Mishkin 2008)

*“Liquidity provision by the central bank **reduces systemic risk** by assuring market participants that, should short-term investors begin to lose confidence, financial institutions will be able to meet the resulting demands for cash without resorting to potentially destabilizing fire sales of assets.”*
(Bernanke 2009)

Measuring Insurance Reasoning

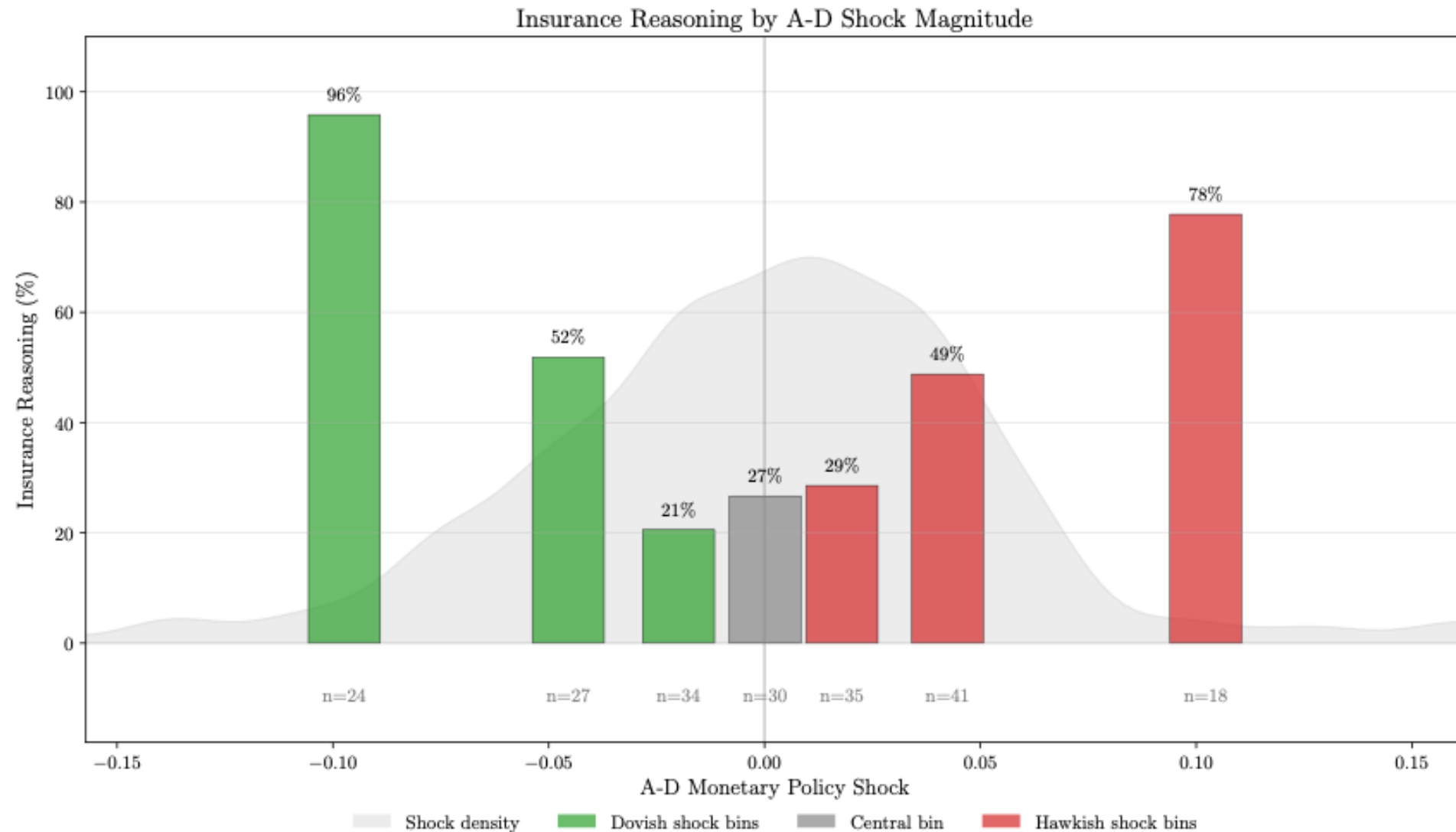
To separate meetings by the presence of insurance motives, we design a new LLM query for all meetings that cover our shock measures (Oct 1982 – Dec 2008).

Collect FOMC2 statements of non-dissenters and require two conditions:

1. Members invoke insurance motive “explicitly argue for taking action to hedge against uncertain future risks even if not the most likely scenario.”
2. Explicit evidence that insurance used as direct motivation for policy choice.

95/206 meetings (46%) have insurance reasoning as motivation for policy.

Insurance Associated with Larger Shocks



BoR Affects Shocks Primarily in Insurance Meetings

	AD Shock			
	(1)	(2)	(3)	(4)
BoR Inflation	0.046*** (0.013)		0.048*** (0.013)	-0.002 (0.010)
BoR Growth	0.019 (0.014)		0.017 (0.015)	-0.048*** (0.010)
Insurance Reasoning		-0.008 (0.008)	-0.008 (0.008)	-0.007 (0.012)
BoR Inflation $\times$ Insurance				0.095*** (0.025)
BoR Growth $\times$ Insurance				0.129*** (0.024)
R2	0.074	0.005	0.079	0.281
Num.Obs.	206	206	206	206

Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Conclusion

AI models do not optimally solve all measurement problems in economics

Traditional text analysis is cheaper, more transparent, and in some cases outperforms AI both in traditional test-set performance and in econometric modeling

But it opens new avenues for narrative analysis particular for subjective, hard-to-measure concepts